



BrightLearn
AI & DATA SKILLS

STUDENT BRIEF · DUE 24 MAY 2026

Retail Sales Data Agent

Your task: build a working Data Agent on Databricks for a retail sales dataset. Write your own agent instructions, test it with real questions, and document the journey.

PLATFORM

Databricks

STEPS

10 required

MARKS

Out of 100

SUBMIT VIA

GitHub + BrightLearn

Build · Test · Submit



00 Your Mission

In this task, you will build a working Data Agent on Databricks that answers business questions about a real retail sales dataset — and you will write the agent's instructions yourself, from scratch.

DEFINITION

The Objective

Build a Retail Sales Data Agent on Databricks using the provided retail sales dataset. Your agent must help a business user, manager, or CEO ask questions about a shop's sales performance and receive useful business insights — in plain language, backed by real data.

DEFINITION

Your Deliverable

A complete project published to GitHub: a working Data Agent in Databricks, your own original agent instructions, ten tested questions, three validated answers, screenshots of every step, and a Word document writing it all up. Submit the GitHub link via the BrightLearn Portal by **Sunday, 24 May 2026**.

★ THE GOLDEN RULE — WRITE YOUR OWN INSTRUCTIONS

Every student must write their own agent instructions. Do not copy them from ChatGPT, your classmates, or any online source. Marks are awarded for original thinking — the rules, exclusions, tone, and edge cases that you decide are right for this dataset. Identical or copied instructions will be flagged and penalised.

Your instructions are the most heavily weighted part of this task. Take time on them.



01 The Retail Sales Dataset

You will be given a CSV file of retail sales transactions. Before you can build a useful agent, you need to understand exactly what is in it and what each column means.

DEFINITION

Columns in the Dataset

Transaction ID · Date · Customer ID · Gender · Age · Product Category · Quantity · Price per Unit · Total Amount. Suggested Databricks table name: **retail_sales_data**.

★ WHAT YOU CAN ANALYSE

- **Sales analysis** — Date, Quantity, Price per Unit, Total Amount.
- **Customer analysis** — Customer ID, Gender, Age.
- **Product analysis** — Product Category, Quantity, Total Amount.
- **Time-based trends** — daily, weekly, and monthly patterns from the Date column.
- **Cross-cuts** — gender × category, age-group × spend, category × month, and so on.

★ BEFORE YOU BUILD — KNOW YOUR DATA

Open the dataset in Databricks first. Look at the column types, sample values, and any missing or unusual entries. The better you understand the raw data, the better your agent instructions will be — because you will know what edge cases to handle, what aggregates make sense, and what business questions are actually answerable.



02 The 10-Step Roadmap

Your project follows ten required steps, end-to-end. The toolkit below is your map — each step earns marks, and each step needs screenshot evidence in your final Word document.





03

Steps 1–3: Set Up Your Data

Three set-up steps before the agent can answer anything. Each step needs its own screenshots.

01

UPLOAD THE DATASET

Upload the provided retail sales CSV file into Databricks. **Screenshot the uploaded dataset, the preview, and the file location.**

TRY

Use the Data tab → "Create Table" → "Upload File". Confirm the file size and row count match the original CSV before continuing.

Dataset file	Preview view	File location
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02

REVIEW THE DATASET

Inspect each column. Decide what it can be used for — sales, customer, or product analysis. **Screenshot the preview, the column list, and any summary view.**

TRY

Run `df.printSchema()` and `df.describe()`. Note missing values, unusual ages, or negative quantities — these are edge cases your instructions must handle.

Columns	Types	Summary stats
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03

PREPARE THE TABLE

Register the dataset as a named table with clear descriptions — the agent reads these to understand the data. **Screenshot the table, its description, and the columns.**

TRY

Use `retail_sales_data`. Add a one-line description on the table and on every column.



04 Step 4: Create the Data Agent

Now you build the agent itself inside Databricks and connect it to the `retail_sales_data` table you just prepared.

★ WHAT TO SET UP

- **Agent name** — something specific, e.g. "Retail Sales Insights Agent". Avoid generic names like "My Agent".
- **Connected dataset** — point the agent at your `retail_sales_data` table.
- **Configuration page** — confirm the model, table access, and any default settings.
- **Test connectivity** — ask the agent one trivial question ("How many rows are in the table?") before moving on.

REQUIRED SCREENSHOTS

· The data agent setup page · The connected dataset/table · The agent name · The configuration page



05

Step 5: Write Your Own Instructions

This is the most heavily weighted single step of the task. Your instructions are how your agent becomes yours — not a generic AI. Write them yourself, from scratch.

★ YOUR INSTRUCTIONS MUST COVER

- **The role** of the agent — what it is, what it does, and for whom.
- **The users** it is helping (business owner? store manager? CEO?).
- **The dataset** it should use, named explicitly (retail_sales_data).
- **The question types** it should answer — sales, customers, products, trends.
- **How to explain insights** — tone, length, structure, the use of numbers.
- **How to handle ambiguity** — what if the question is unclear or the data is missing?
- **How to avoid making things up** — no fabricated numbers, no invented categories.
- **How to give recommendations** — when to suggest actions, and when to stop at the facts.

! DO NOT COPY

Do not copy your instructions from ChatGPT, classmates, online tutorials, or this brief. **Marks are awarded for original thinking.** Identical or copied instructions will be flagged and penalised. Write them yourself, in your own words.

REQUIRED SCREENSHOTS

- Your full agent instructions (legible)
- The page where you entered them
- The saved agent configuration confirming the instructions are live



06

Step 6: Test with 10 Questions

Once your agent has instructions, exercise it. You must ask at least 10 business questions that you have written yourself — they cannot be copied from this brief.

EXAMPLE AREAS — PICK YOUR OWN ANGLE INSIDE EACH

Total sales · Sales by product category · Sales by gender · Sales by age group
Best-performing category · Lowest-performing category · Monthly sales trends
Average transaction value · Customer buying behaviour · Business recommendations

★ HOW TO WRITE GOOD TEST QUESTIONS

Mix easy and hard. Some questions should have one clear answer ("What is total revenue?"). Others should force the agent to combine fields ("Which product category sells best to customers under 30?"). At least one should be deliberately tricky or ambiguous — that is how you find out whether your instructions handle uncertainty well.

REQUIRED SCREENSHOTS

For every one of the 10 questions: the question you typed, the agent's answer, and any chart, table, or summary it produced.



07

Step 7: Validate the Answers

An agent that sounds confident is not the same as an agent that is correct. You must independently check at least three of its answers against the source data.

★ WAYS TO VALIDATE AN ANSWER

- **Dataset summaries** — re-run `df.describe()` or `df.groupby().agg()` to get the same number yourself.
- **Databricks table previews** — sort and filter the table manually to confirm the result.
- **Charts** — visualise the data and check the agent's claim against the visual pattern.
- **Aggregated SQL** — write your own SQL query and compare the result.
- **Cross-check questions** — ask the agent the same thing two different ways and check the answers match.

REQUIRED FOR EACH OF YOUR 3 VALIDATIONS

- The agent's original answer
- The supporting evidence you used to check it
- A short, honest written explanation of whether the answer is correct, partially correct, or wrong.

★ HONESTY EARNS MARKS

If an answer is wrong, say so — and explain why you think it went wrong. Finding a real mistake in your own agent and writing about it cleanly is worth more than three confidently asserted right answers. This is what data professionals actually do.



08

Step 8: The Write-Up Document

All your work goes into a single Word document. Suggested title: Building a Retail Sales Data Agent on Databricks.

★ YOUR WORD DOCUMENT MUST CONTAIN

- **Title page** — title, your name, your student number, date.
- **Introduction & Objective** — what the project is and why you did it.
- **Dataset description** — what is in retail_sales_data and what each column means.
- **Screenshots** — upload, review, table prep, agent setup, instructions.
- **Your agent instructions in full** — the exact text you entered into Databricks.
- **The 10 questions** — and the agent's answer to each, with screenshots.
- **Validation of 3 answers** — the evidence and your written verdict on each.
- **Key insights** — what your agent revealed about the shop's sales.
- **Business recommendations** — what the business should do next, in your own words.
- **Conclusion** — what worked, what was hard, what you would change.
- **GitHub repository link** — at the end of the document.



09

Steps 9 & 10: GitHub & Submit

Your work is not complete until it is on GitHub and the link is in the BrightLearn Portal. Both steps are required.

09

PUSH TO GITHUB

Create a public GitHub repository for this project. Upload the Word document, the dataset (if permitted), and a README.md.

TRY

Suggested README sections: Project title · Project objective · Tools used · Dataset overview · Steps followed · Your agent instructions · Sample questions tested · Key insights · Conclusion.

Word doc**Dataset****README.md**

10

SUBMIT ON BRIGHTLEARN PORTAL

Submit the link to your GitHub repository on the BrightLearn Portal before the deadline.

TRY

Open your GitHub repo in an incognito window before submitting — it must be readable without a login. Private repos cannot be marked.

Public repo**Working link****By 24 May 2026****! LATE OR MISSING
SUBMISSIONS**

Submissions after Sunday 24 May 2026 **will not be marked**. A working GitHub link in the Portal by the deadline is the only valid submission. Don't email it, don't message it — submit it through the Portal.



10

Marking & Final Checklist

This is exactly how your work will be marked. Use the checklist on the right as the very last thing you do before pressing submit.

MARKING GUIDE · TOTAL OUT OF 100

Dataset uploaded correctly in Databricks	10
Dataset reviewed and understood	10
Dataset prepared for the agent	10
Data agent created and connected to the dataset	20
Student-created agent instructions (original work)	20
At least 10 original questions tested	10
At least 3 answers validated	10
Screenshots included throughout the document	5
GitHub repository submitted correctly via the Portal	5

★ FINAL SUBMISSION CHECKLIST

- Dataset uploaded, reviewed, and prepared as retail_sales_data
- Data agent created and connected to the table
- Your own original agent instructions written and saved
- Agent tested with at least 10 original questions
- At least 3 answers independently validated
- Screenshots captured for every step
- Word document compiled and reviewed
- Repository pushed to public GitHub with a README
- GitHub link submitted on the BrightLearn Portal by Sunday, 24 May 2026



Build it. Test it. Ship it.

This task is your first end-to-end Data Agent build. Take it seriously, write instructions you are proud of, and submit by the deadline.

Sunday, 24 May 2026 · BrightLearn Portal.

★ BRIGHTLEARN · STUDENT TASK BRIEF ★